



Introduction to Medical Virology: Understanding Viruses and Their structural properties

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Exploring viruses and their composition

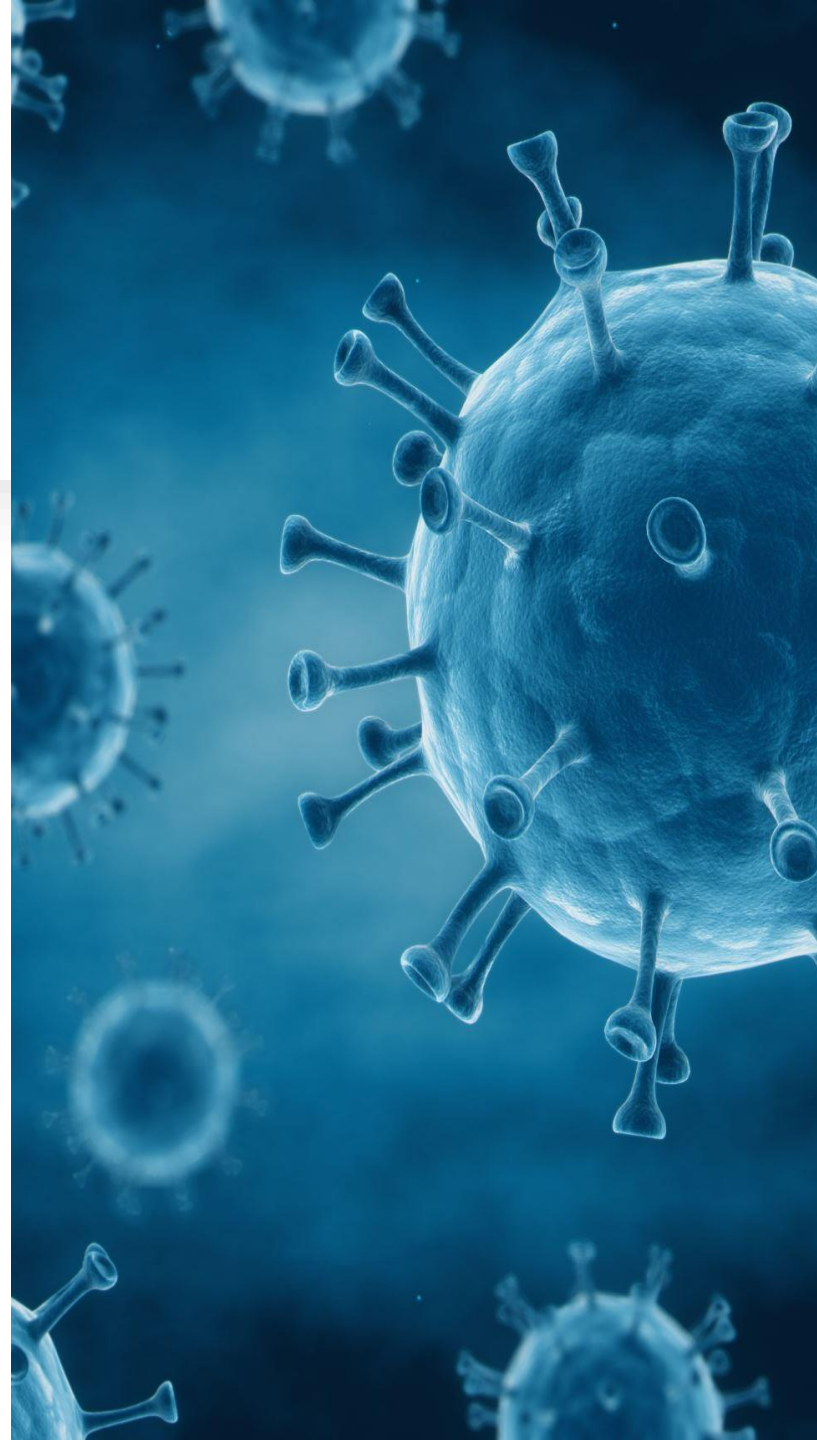


Objectives

- Definition of Viruses
- Structure of viruses
- Viral symmetry
- Classification of viruses

General property of viruses

- Viruses are not cells; they are not capable of independent replication, can synthesize neither their own energy nor their own proteins, and are too small to be seen in the light microscope
- Viruses are particles composed of an internal core containing *either*
- Viruses must reproduce (replicate) within cells
- Viruses replicate in a manner different from that of cells



Viruses

Viruses are the smallest infectious agents (ranging from about **20 to 300 nm** in diameter) and contain **only one kind of nucleic acid (RNA or DNA)** as their genome. The nucleic acid is **encased in a protein shell**, which may be surrounded **by a lipid-containing membrane**. The entire infectious unit is termed a *virion*.

Comparison of viruses and cells

Property	Viruses	Cells
Type of nucleic acid	DNA or RNA but not both	DNA and RNA
Proteins	Few	Many
Lipoprotein membrane	Envelope present in some viruses	Cell membrane present in all cells
Ribosomes	Absent ¹	Present
Mitochondria	Absent	Present in eukaryotic cells but not in prokaryotic cells
Enzymes	None or few	Many
Multiplication by binary fission or mitosis	No	Yes

¹Arenaviruses have a few nonfunctional ribosomes.

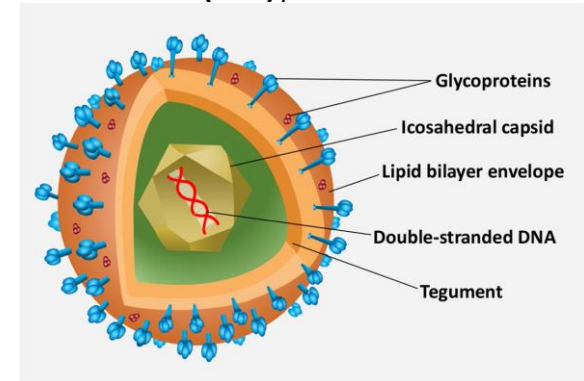
Viruses: Structure

➤ **Genome:**

- A core of DNA or RNA, single-stranded (ss) or double stranded (ds), circular or linear

➤ **Capsid:** protein coat surrounding the genome

- Provides structural symmetry
- Participates in attachment to susceptible host
- Facilitates transfer of viral nucleic acid in to host cell
- Protects the viral genome from nucleases in blood stream

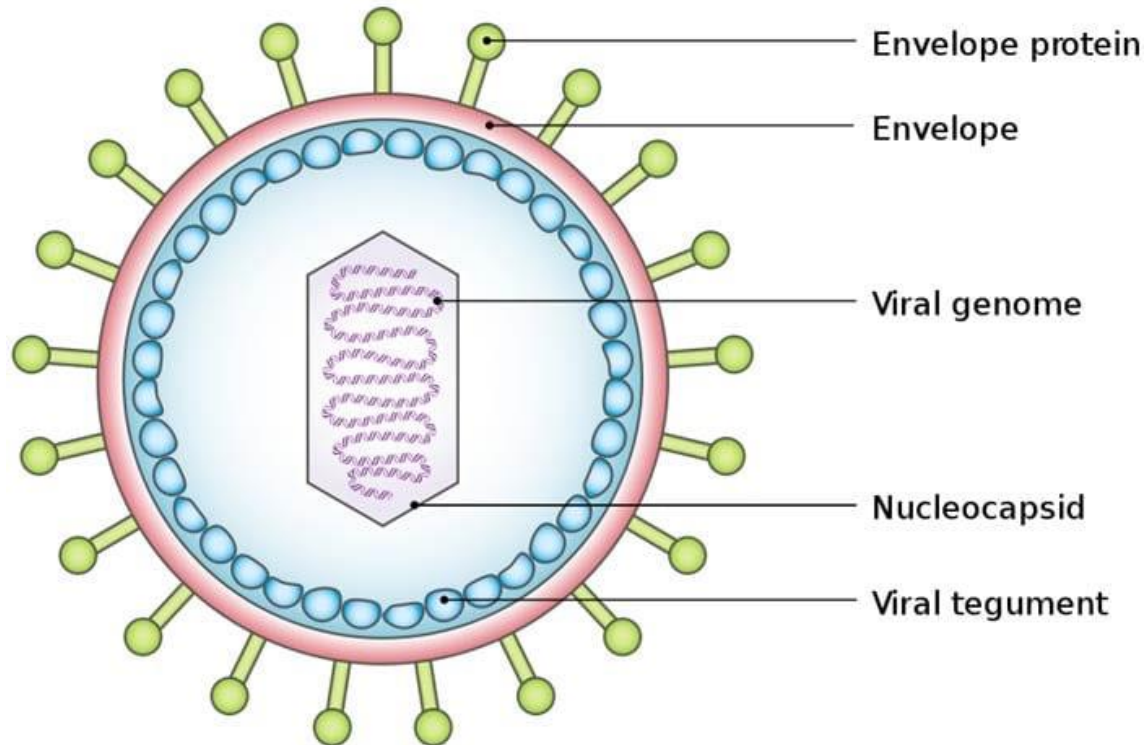


➤ **Other Viral Proteins (tegument)**

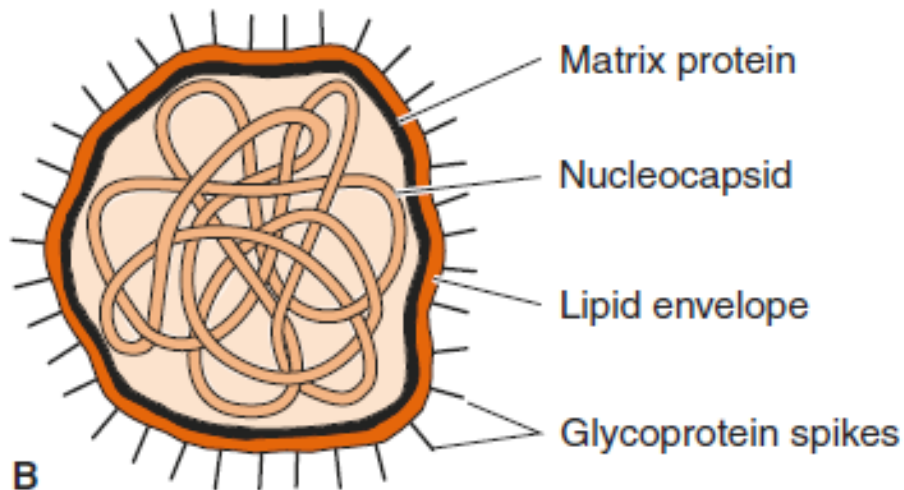
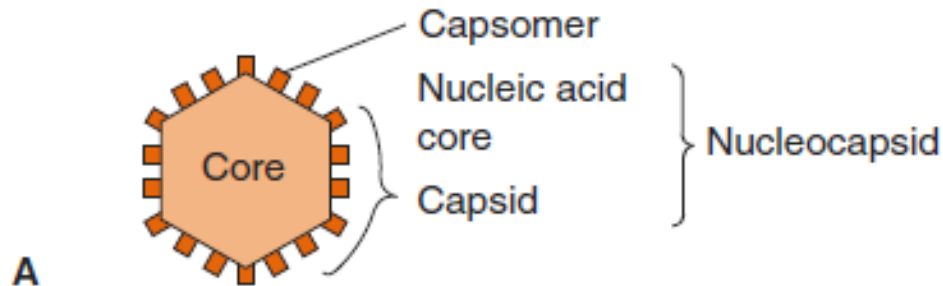
➤ **Envelope:**

- A lipoprotein surrounding the capsid in some viruses
- Virus-encoded glycoproteins are exposed on the surface of the envelope

Structure of virus



A: Nonenveloped virus with an icosahedral nucleocapsid.



B: Enveloped virus with a helical nucleocapsid

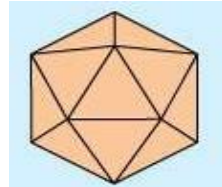
Viral Structure

- The RNA can exist either as a single molecule or in several pieces.
- Almost all viruses contain only a single copy of their genome (i.e., they are **haploid**).
- Viral nucleocapsids have two forms of symmetry
- Viral proteins serve several important functions
 - Capsid proteins **protect the genome**
 - **Mediate the attachment**
 - **Species and organ specificity**
 - Induce neutralizing antibody
 - Is used to describe a subcategory of a virus “**serotype**”

Viruses: Symmetry (Shapes)

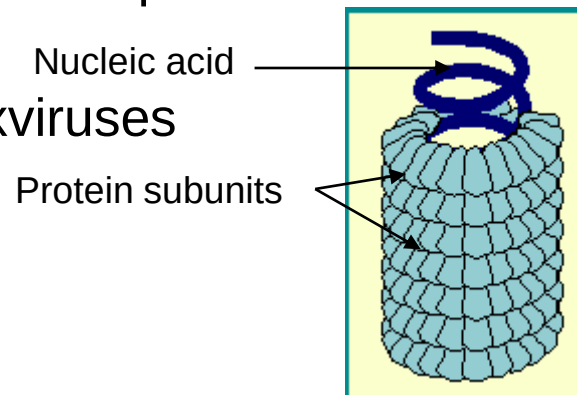
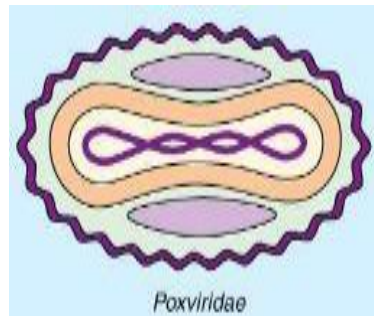


1. **Cubic Symmetry (Icosahedral)** Composed of 20 faces .e.g. Herpesviruses , Adenoviruses.

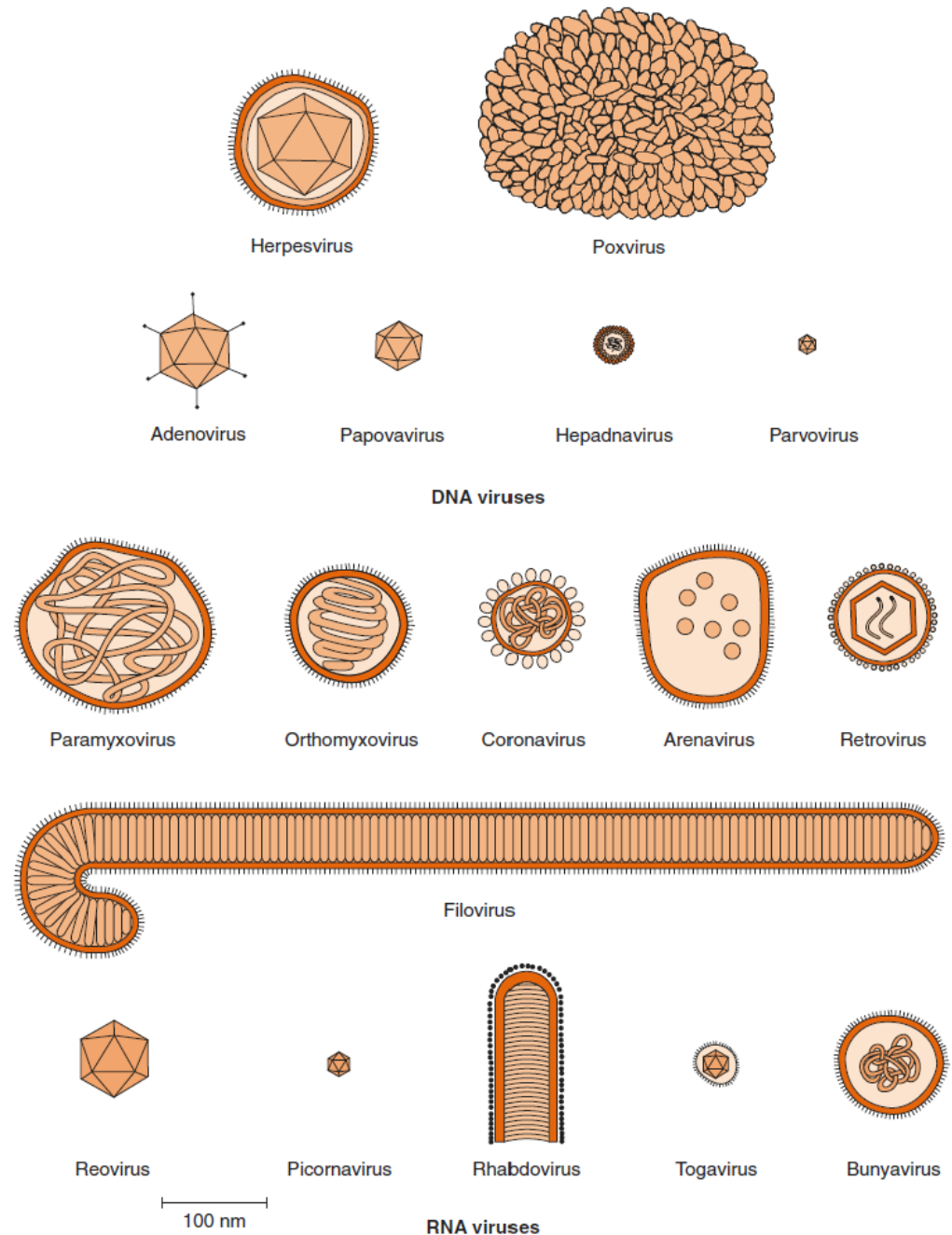


2. **Helical Symmetry** In which the capsomeres are arranged in a hollow coil that appears rod shaped.

3. **Complex structures, e.g. Poxviruses**



Shapes and sizes of Medically important viruses



Structure of viruses

There are four exceptions to the **typical virus**
(Atypical Virus-like Agents)

1. **Defective viruses** Are composed of viral nucleic acid and proteins but cannot replicate without a 'helper' virus
2. **Pseudovirions** contain host cell DNA instead of viral DNA within the capsid
3. **Viroids** consist solely of a single molecule of circular RNA without a protein coat or envelope.
4. **Prions** are infectious particles that are composed **solely of protein**

Virus Replication Cycle

Attachment to Host Cell

The first stage involves the virus attaching itself to specific receptors on the surface of the host cell.

Penetration

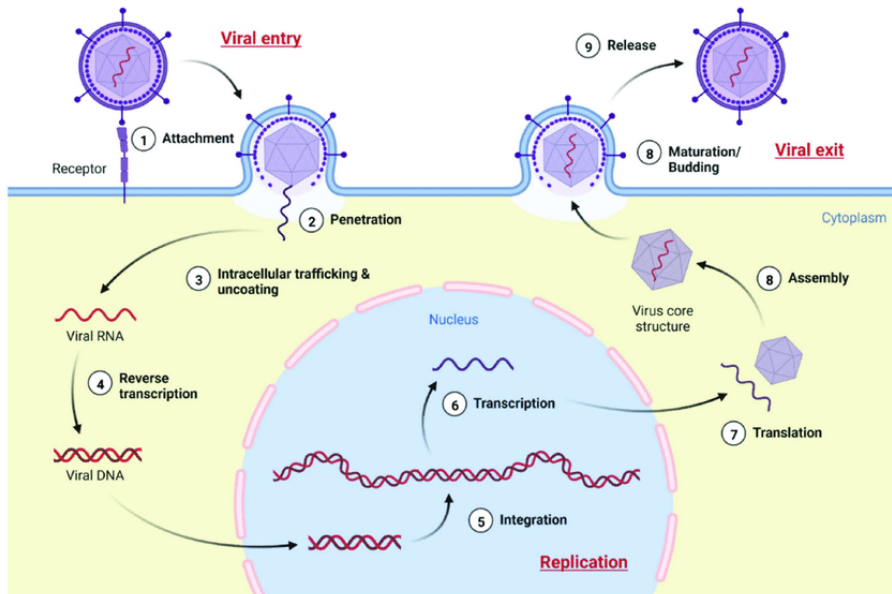
Following attachment, the virus penetrates the host cell membrane to enter the cytoplasm.

Replication

Inside the host cell, the virus replicates its genetic material and produces new viral components.

Assembly and Release

New viral particles are assembled and released from the host cell, ready to infect new cells.

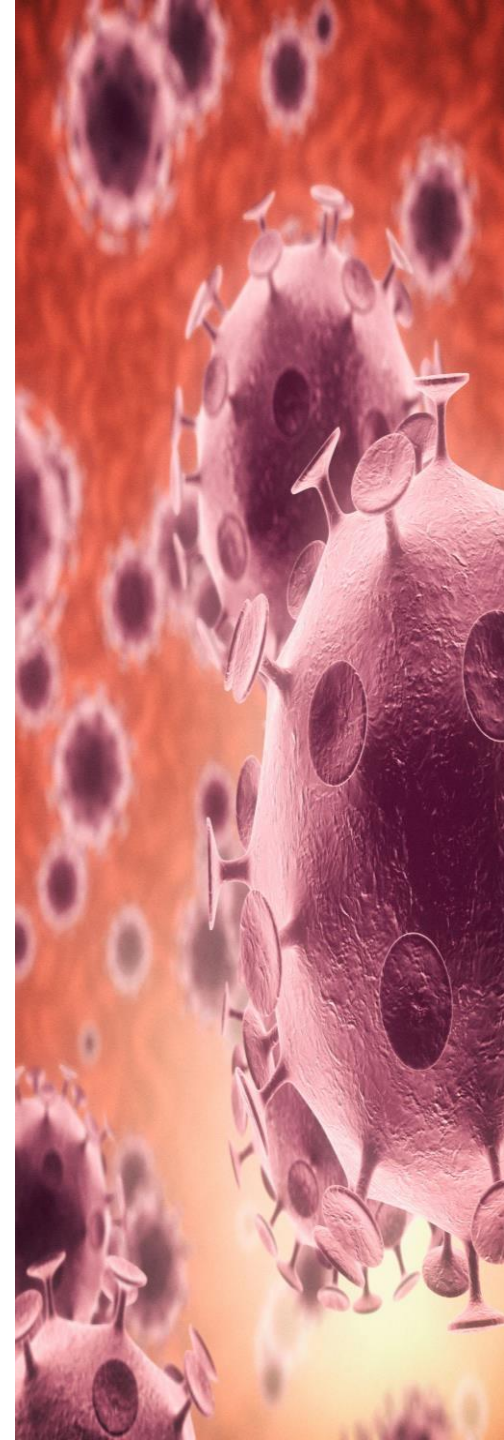


Classification of Viruses

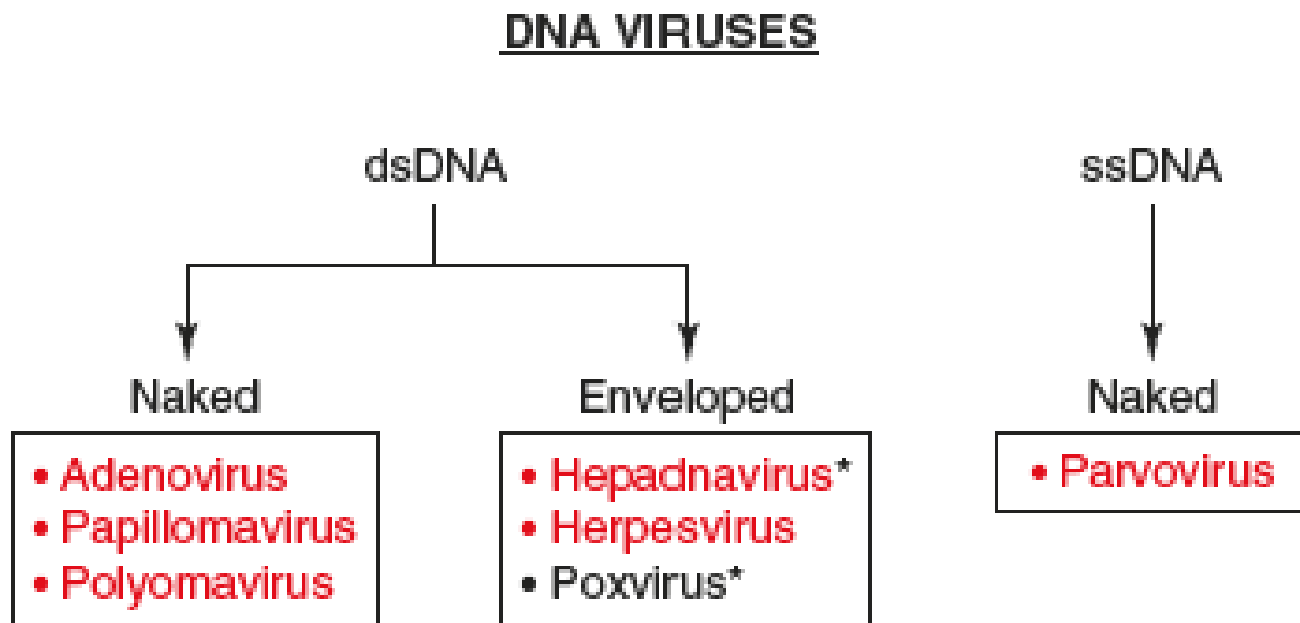


PRINCIPLES OF CLASSIFICATION

- Classification of viruses is based on chemical and morphologic criteria
- The two major components of the virus used in classification are
 - The nucleic acid(DNA or RNA & ss or ds)
 - The capsid
 - Its size and symmetry (Icosahedral or Helical)
 - Enveloped. (Enveloped or Naked)



Classification scheme for DNA viruses.



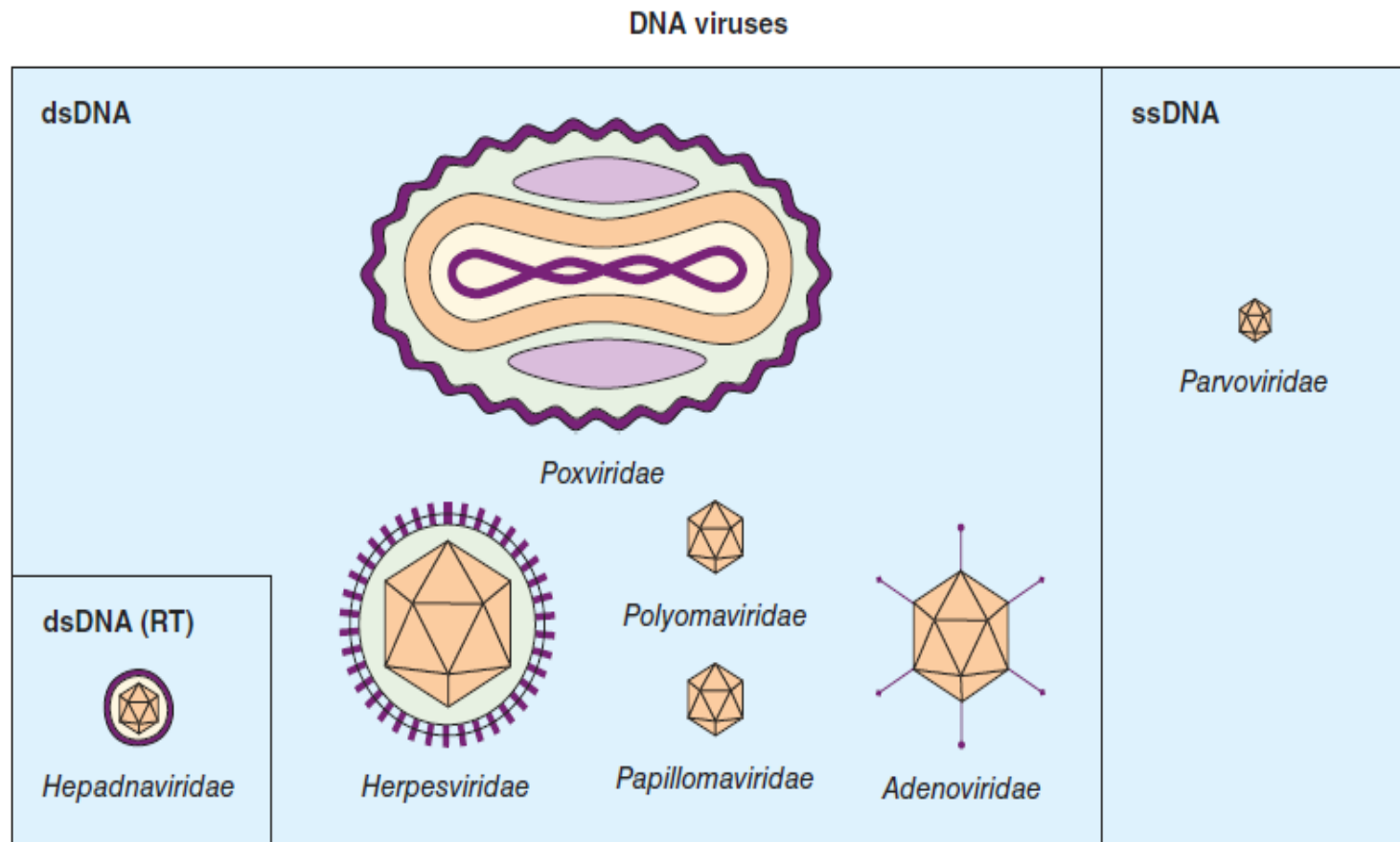
Red = icosahedral symmetry
(Poxvirus has complex symmetry)

*Viruses labeled with asterisk
have a polymerase in the virion

DNA VIRUSES

- The four **naked** (i.e., nonenveloped) icosahedral virus families—
 - Parvoviruses,
 - Polyomaviruses,
 - Papillomaviruses, and
 - Adenoviruses
- The three **enveloped** families. The
 - Hepadnavirus
 - Herpesviruses
 - Poxviruses

Properties of DNA Viruses



Property of DNA viruses

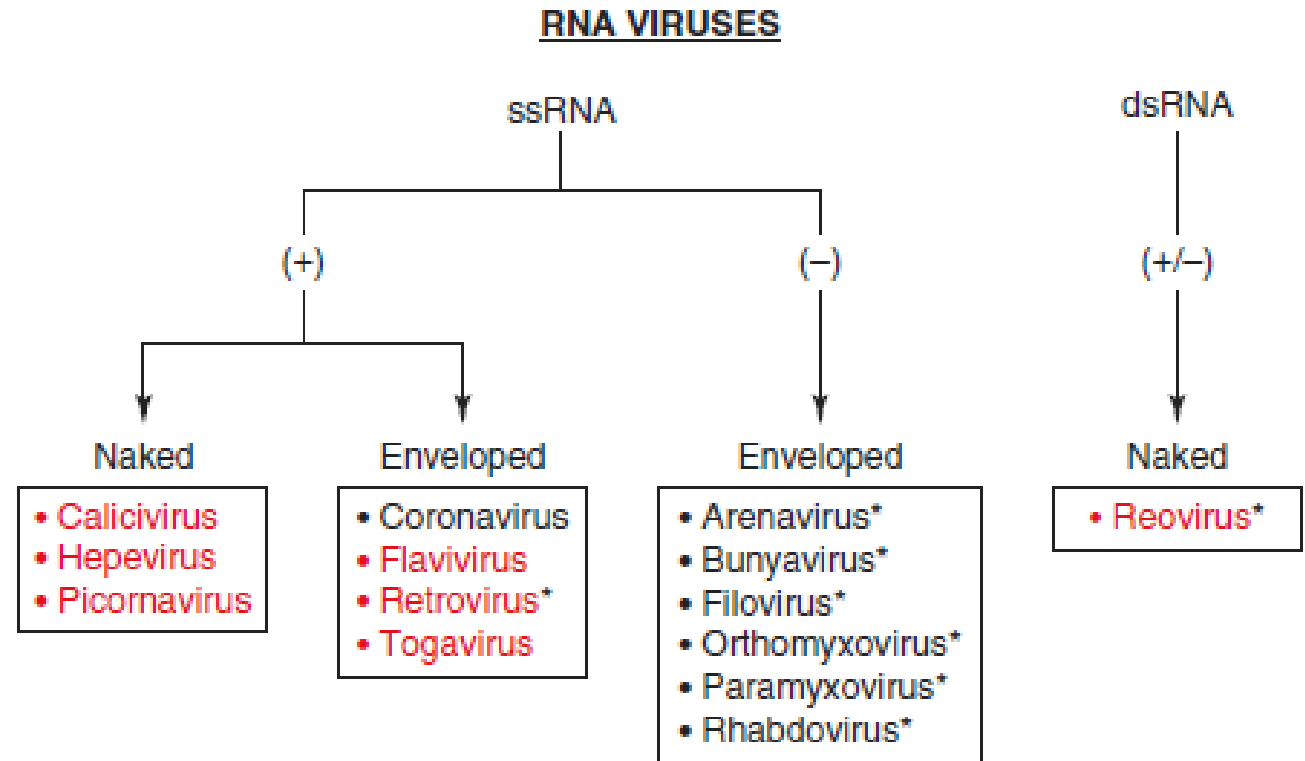
Virus Family	Envelope Present	Capsid Symmetry	Virion Size (nm)	DNA MW ($\times 10^6$)	DNA Structure ¹	Medically Important Viruses
Parvovirus	No	Icosahedral	22	2	ss linear	B19 virus
Polyomavirus	No	Icosahedral	45	3	ds circular, supercoiled	JC virus, BK virus
Papillomavirus	No	Icosahedral	55	5	ds circular, supercoiled	Human papillomavirus
Adenovirus	No	Icosahedral	75	23	ds linear	Adenovirus
Hepadnavirus	Yes	Icosahedral	42	1.5	ds incomplete circular	Hepatitis B virus
Herpesvirus	Yes	Icosahedral	100 ²	100–150	ds linear	Herpes simplex virus, varicella-zoster virus, cytomegalovirus, Epstein–Barr virus
Poxvirus	Yes	Complex	250 $\times$ 400	125–185	ds linear	Smallpox virus, molluscum contagiosum virus

¹ss, single-stranded; ds, double-stranded.

²The herpesvirus nucleocapsid is 100 nm, but the envelope varies in size. The entire virus can be as large as 200 nm in diameter.

Classification scheme for RNA viruses.

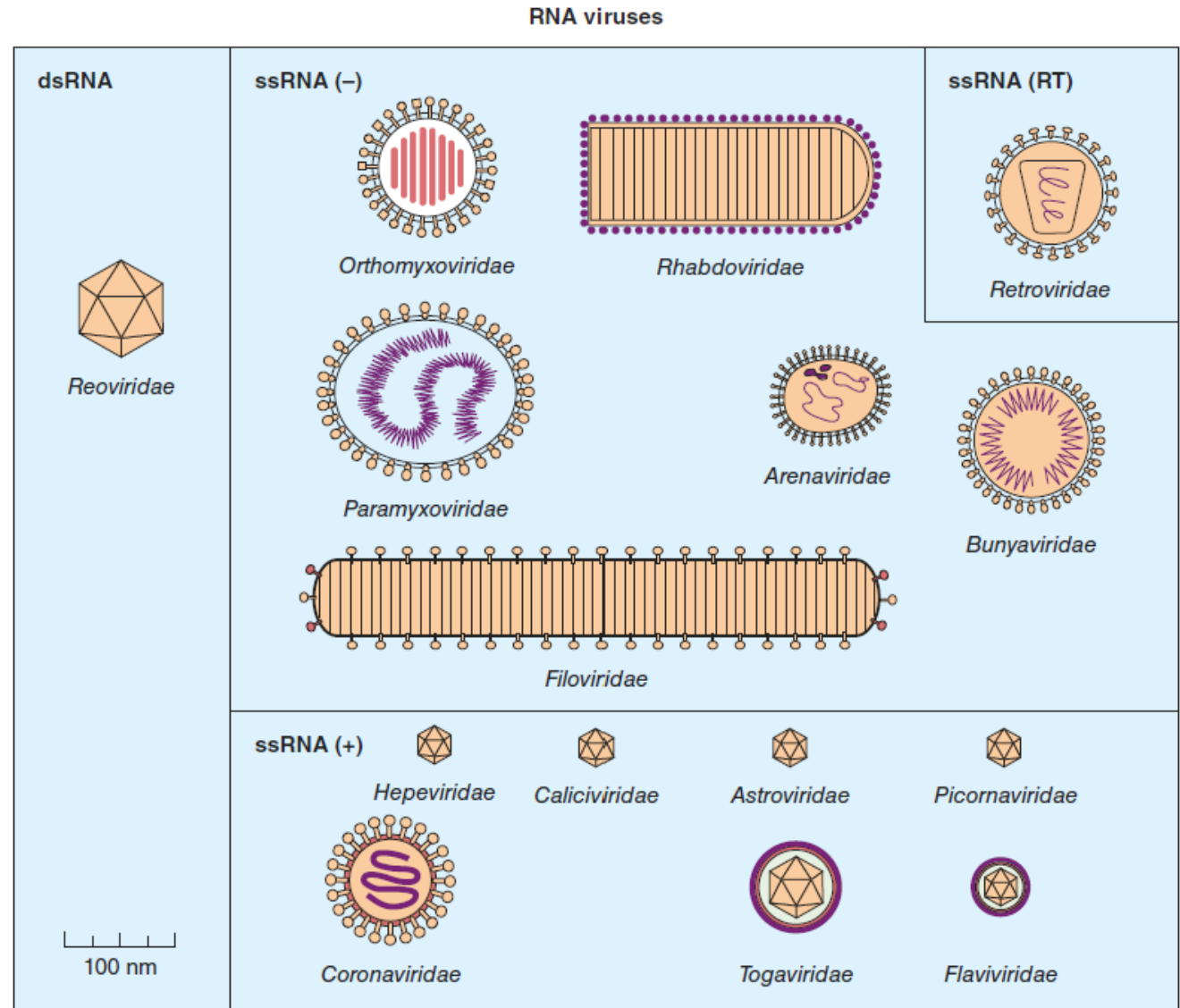
The 14 families of RNA viruses



Red = icosahedral symmetry
(All others have helical symmetry)

* Viruses labeled with asterisk
have a polymerase in the virion

Properties of RNA Viruses



Property of RNA viruses

Virus Family	Envelope Present	Capsid Symmetry	Particle Size (nm)	RNA MW ($\times 10^6$)	RNA Structure ¹	Medically Important Viruses
Picornavirus	No	Icosahedral	28	2.5	ss linear, nonsegmented, positive polarity	Poliovirus, rhinovirus, hepatitis A virus
Hepevirus	No	Icosahedral	30	2.5	ss linear, nonsegmented, positive polarity	Hepatitis E virus
Calicivirus	No	Icosahedral	38	2.7	ss linear, nonsegmented, positive polarity	Norovirus
Reovirus	No	Icosahedral	75	15	ds linear, 10 or 11 segments	Rotavirus
Flavivirus	Yes	Icosahedral	45	4	ss linear, nonsegmented, positive polarity	Yellow fever virus, dengue virus, West Nile virus, hepatitis C virus, Zika virus
Togavirus	Yes	Icosahedral	60	4	ss linear, nonsegmented, positive polarity	Rubella virus
Retrovirus	Yes	Icosahedral	100	7 ²	ss linear, 2 identical strands (diploid), positive polarity	HIV, human T-cell leukemia virus
Orthomyxovirus	Yes	Helical	80–120	4	ss linear, 8 segments, negative polarity	Influenza virus
Paramyxovirus	Yes	Helical	150	6	ss linear, nonsegmented, negative polarity	Measles virus, mumps virus, respiratory syncytial virus
Rhabdovirus	Yes	Helical	75 $\times$ 180	4	ss linear, nonsegmented, negative polarity	Rabies virus
Filovirus	Yes	Helical	80 ³	4	ss linear, nonsegmented, negative polarity	Ebola virus, Marburg virus
Coronavirus	Yes	Helical	100	10	ss linear, nonsegmented, positive polarity	Coronavirus
Arenavirus	Yes	Helical	80–130	5	ss circular, 2 segments with cohesive ends, negative polarity	Lassa fever virus, lymphocytic choriomeningitis virus
Bunyavirus	Yes	Helical	100	5	ss circular, 3 segments with cohesive ends, negative polarity	California encephalitis virus, hantavirus
Deltavirus	Yes	Uncertain ⁴	37	0.5	ss circular, closed circle, negative polarity	Hepatitis delta virus

¹ss, single-stranded; ds, double-stranded.

²Retrovirus genome RNA contains two identical molecules, each with a molecular weight (MW) of 3.5×10^6 .

³Particles are 80 nm wide but can be thousands of nanometers long.

⁴The nucleocapsid appears spherical but its symmetry is unknown.

Summary

- Viruses contain **either DNA or RNA, but not both**
- All viruses have a **protein coat called a capsid**
- Some RNA viruses, have a **segmented genome**
- Viral surface proteins mediate **attachment to host cell receptors**
- Some viruses produce **antigenic variants** of their surface proteins
- **Prions** are infectious particles composed **entirely of protein**
- Prion proteins are **encoded by a cellular gene**

Summary

- The classification of viruses is based primarily on **the genome** and whether the virus **has an envelope**.
- Parvoviruses have single-stranded DNA, whereas all the other families of DNA viruses have double-stranded DNA
- Reoviruses have double-stranded RNA; all the other families of RNA viruses have single-stranded RNA
- Reoviruses and influenza viruses have segmented RNA; all the other families of RNA viruses have nonsegmented RNA



References

- **Medical Microbiology**, Jawetz Melnick & Adelbergs (Lange)
- **CDC** centre for disease transmission and prevention
- **Davidson's Principles and Practice of Medicine**,

Thank you

